SIH 2026 · PS 26052 (DRDO) · Team ZEIT v.1.0 (HW2026106)

ZEIT System Architecture

The final, locked architecture — both lanes, every step, and why each step exists

Version	2.0 — draft for team review
Date	17 September 2026
Status	Designed, not built. Every claim carries a status tag
Companion document	ZEIT Hardware (hardware IDs H1–H24 are used throughout this document)
Supersedes	Architecture v1 (Raspberry Pi 5 design) in handbook/index.html and docs/01-system/architecture.md

Five blank revision pages are provided at the end of this document for your changes.

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1. How to use this document

This document describes what the system does, step by step. Its companion, ZEIT Hardware, lists every physical part, port and wire. Both follow docs/01-system/architecture.md and synthetic_generator/HARDWARE.md in the repository, and share one set of IDs:

ID	Meaning
A1 – A11	Lane A steps (the wearer's ears), on the ADAU1787 DSP
R1 – R5	Lane B receive steps (the teammate's voice to the ears), on the Jetson
B1 – B11	Lane B transmit steps (the wearer's voice to the radio), on the Jetson
T1 – T8	Neural network training and deployment (offline)
H1 – H24	Hardware items in the companion document
Mic 1, 2L, 2R, 3L, 3R, 4	The six microphones (section 4)

Status tags

Tag	Meaning
VERIFIED	Checked against a published source (datasheet, standard, manufacturer page)
MEASURED	We ran it ourselves and have the number
ESTIMATE	Engineering arithmetic; not yet tested
UNVERIFIED	A risk. Must be checked before relying on it

2. The problem and the physics

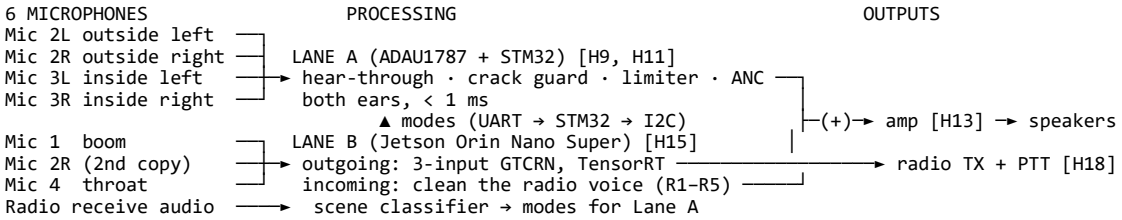
> **IN PLAIN WORDS** A soldier talks on the radio while guns fire nearby. The soldier's ears are being damaged, yet the soldier must still hear what is around them. The listener at the other end should hear the order, not the gunfire — and the soldier must understand the reply. Two very different jobs: one must react in under a thousandth of a second, the other has tens of milliseconds.

PS 26052: an AI/ML-enabled adaptive noise cancellation system for stationary, non-stationary and impulsive defence noise, with high speech intelligibility and real-time performance on embedded hardware (DSPs or AI SoCs such as NVIDIA Jetson, with ONNX / TensorRT), integrated with primary + reference microphones and headphones / communication units. Targets: SNR > 15 dB, STOI > 0.85, PESQ > 2.5, low latency.

Fact	Number	Consequence	Status
Weapons exceed safe limits	Most weapons 150–180+ dBP; MIL-STD-1474E limit at the ear 140 dBP	10–40 dB must be removed before the eardrum, even with a flat battery	VERIFIED
Consumer ANC is not built for gunfire	Apple: Hearing Protection not suitable for gunfire or above 110 dBA	Consumer ANC cannot be called gunfire protection	VERIFIED
Sound crosses an earcup very fast	≈ 0.3–0.6 ms	Anti-noise must be faster; only a dedicated audio DSP qualifies	ESTIMATE
A bullet's crack arrives before the blast	Shooter at 30 m: crack 45.2 ms, blast 87.5 ms, gap 42.3 ms	A free warning the system uses	ESTIMATE
Radio voice tolerates delay	ITU-T G.114: ≤ 150 ms one-way is transparent	The AI has tens of milliseconds; the ear path does not	VERIFIED

3. Two lanes

> **IN PLAIN WORDS** Lane A lives in the headset electronics and looks after the wearer's ears: quiet sounds in, loud ones squashed, engine hum cancelled, in both ears. Lane B lives on the Jetson and looks after the voice: it cleans the wearer's voice for the radio with AI, and cleans the teammate's incoming voice before it reaches the ears. The Jetson also tells Lane A which mode to be in, but Lane A never waits for the Jetson to protect the ear.



	Lane A — the ear	**Lane B — the voice**
Serves	Wearer's hearing and awareness	Listener (outgoing voice) and the wearer's understanding of the reply
Runs on	ADAU1787 DSP (H9) + STM32 (H11)	Jetson Orin Nano Super (H15)
Time budget	< 1 ms ANC; ≤ 5 ms hear-through	≤ 150 ms mouth-to-ear; design ≈ 70–85 ms
AI	No per-sample AI; mode chosen by Lane B	GTCRN neural network + classical blocks
Inputs	Mic 2L, 2R, 3L, 3R	Mic 1, reference (Mic 2R or 2L), Mic 4; radio receive
Outputs	Both speakers (anti-noise + hear-through)	Radio TX + PTT; incoming voice to both speakers
Measured by	Noise at a dummy ear; peak vs 140 dBP; 82 dB ceiling	SNR, STOI, PESQ, SI-SNR, IRT, WLPS, latency, watts
If it fails	Passive earmuff still protects	Bypass: raw boom mic to radio

4. The headset: six microphones

> **IN PLAIN WORDS** A 3M PELTOR Optime III earmuff (SNR 35 dB), modified by us: two speakers and five microphones fitted, plus a throat mic. Three mics carry the voice or the noise reference to the AI; four serve the ears. Mic 2R does both.

Why an earmuff and not an aviation headset: no electronics of its own, so every dB of improvement between system-off and system-on is demonstrably ours; more passive protection (SNR 35 vs ~24 dB); cheap to modify and to replace; our own left and right speakers.

Mic	Position	Hears	Job / lane	Hardware
Mic 1	Boom, left, 2–5 cm from mouth	Voice + noise	Voice to clean — Lane B, AI input 0	H1-B
Mic 4	Throat (laryngophone)	Voice as vibration; little airborne noise	Voice guide — Lane B, AI input 2	H4
Mic 2R	Outside right cup, opposite boom	The world, least voice	AI noise reference (input 1); right ANC; right hear-through	H2
Mic 2L	Outside left cup	The world, left (some voice)	Left ANC; left hear-through	H2
Mic 3R	Inside right cup	What reaches the right ear	Right ANC error	H3
Mic 3L	Inside left cup	What reaches the left ear	Left ANC error	H3

AI inputs — locked order

AI input	Mic	Interface input	Why the model needs it
0	Mic 1 boom	UMC404HD in 1	The signal that is cleaned; full voice bandwidth, full of gunfire
1	Reference: Mic 2R (Mic 2L for left-handed)	UMC404HD in 2	The noise with almost no voice; tells noise from voice
2	Mic 4 throat	UMC404HD in 3	Voice gunfire cannot reach; shows where the words are during a blast

> **RULE** The order must be identical in training, in the exported model and in the cabling. Changing it after training silently ruins the model.

Placement rules

- **Boom on the left.** A right-handed soldier shoulders the rifle on the right; 3M's ComTac VIII mounts its speech mic on the left cup (VERIFIED).

- **Reference on the far side from the mouth.** Whatever the reference hears is learned as noise; if it hears the voice, the voice is removed. Checked by the voice-leak test (reference ≥ 25 dB quieter than Mic 1).
- **Handedness switch.** For a left-handed soldier the boom moves right and a switch on the preamp board makes Mic 2L the reference. No retraining.
- **Why not Mic 2L or Mic 3 into the AI:** Mic 2L hears too much voice; Mic 3L / 3R hear our own speakers, not the world.

5. Protection chain (passive first)

IN PLAIN WORDS Three walls stand between a gunshot and the eardrum: the sealed earmuff (no power needed), an optional foam earplug for live fire, and the electronics — which may only make things quieter or safer, never louder than 82 dB. If the battery dies the first two still stand. Protector ratings (EN 352) are measured with the electronics off (3M ComTac VIII datasheet, VERIFIED).

TWO MEANINGS OF SNR On a hearing protector, SNR = Single Number Rating (dB removed). In the speech metrics, SNR = signal-to-noise ratio of the transmitted voice (PS target > 15 dB). Never mix them.

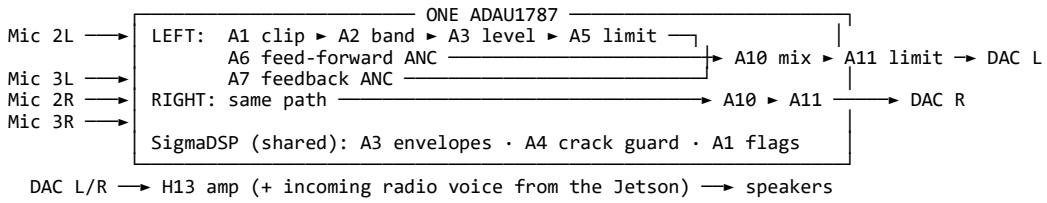
Stage	Level at ear	Status
Rifle shot at the head	~155 dB	ESTIMATE
After the sealed earmuff (SNR 35)	~125–130 dB	ESTIMATE — peaks differ from the steady-noise rating; measure on the dummy head
After earmuff + foam earplug	~115–120 dB	Double protection +5–10 dB (OSHA rule of thumb +5 dB) — VERIFIED rule, ESTIMATE result
MIL-STD-1474E limit	140 dBP	VERIFIED
Anything the electronics play	≤ 82 dB	Design rule, enforced in two places (section 6)

SCOPE Heavy weapons (180+ dBP) exceed what any headset in this class can bring below 140 dBP. The protection claim is for small arms at a distance.

6. Lane A — the ear: steps A1–A11

> **IN PLAIN WORDS** The outside mics listen to the world. Quiet sounds are turned up so the earmuff does not make you deaf to your surroundings; loud sounds are turned down and nothing goes above 82 dB. When the DSP hears a bullet's crack it lowers the volume before the much louder blast arrives. In both ears, ANC cancels engine and rotor rumble. One ADAU1787 does all of this for both ears; the same path runs twice, left and right.

ADAU1787 port	Signal
ADC 1	Mic 2L
ADC 2	Mic 2R (second buffered copy)
ADC 3	Mic 3L
ADC 4	Mic 3R
DAC L / R	Left / right ear → TPA6130A2 amp (H13)
I²C	Control and program load from the STM32 (via I²C level shifter H12a, PCA9306)
Serial port (I²S/TDM)	Mic data to the STM32 — adaptive ANC phase only
USB	SigmaStudio on a laptop — development only



> **GATE 2** The pin count matches stereo hybrid ANC exactly. Still to confirm: that the fast DSP path has capacity for both ears' ANC filters at once. If not, ANC filters are shortened; hear-through and limiters are unaffected.

A1 Clip detection

Purpose (plain)	Notice when an outside mic is overloaded, so nothing downstream trusts a destroyed signal.
What it does	Flags samples near full scale on Mic 2L / 2R. The flag fires the crack guard (A4) and freezes ANC adaptation (A8).
Runs on	SigmaDSP
Hardware used	H2, H6, H9
Timing	Within a few samples
Status	Designed. Mic 2 overload point 135 dB SPL (VERIFIED) — a nearby crack will clip it

A2 Hear-through band filter

Purpose (plain)	Keep what a soldier needs to hear (voices, footsteps, vehicles), drop rumble and hiss.
What it does	Biquad band-pass ≈ 150 Hz – 6 kHz, each ear.
Runs on	FastDSP
Hardware used	H9
Timing	Microseconds
Status	Designed; biquads are native (VERIFIED)

A3 Level-dependent gain

Purpose (plain)	Turn quiet sounds up, loud sounds down.
What it does	Upward expander below ≈ 50 dB (up to +12 dB), unity in the middle, compressor above ≈ 70 dB; envelope on SigmaDSP, gain on FastDSP; 5–50 ms gain changes. Settings per mode.
Runs on	SigmaDSP + FastDSP
Hardware used	H9
Timing	< 1 ms signal delay
Status	Designed; copied from Peltor ComTac VIII level-dependent mode (VERIFIED)

A4 Crack guard

Purpose (plain)	Hear a bullet's crack and lower hear-through before the blast arrives ≈ 40 ms later.
What it does	Fast peak + high-frequency detector on Mic 2L / 2R, also fired by A1. Hear-through in BOTH ears drops 20–30 dB instantly, holds ≈ 300 ms, releases over ≈ 100 ms; mode → COMBAT.
Runs on	SigmaDSP
Hardware used	H9
Timing	A few samples; blast ≈ 42 ms later (ESTIMATE)
Status	Designed — our own idea

A5 Hear-through limiter

Purpose (plain)	A hard ceiling on the hear-through.
What it does	Peak limiter calibrated on the mannequin (H22) to the 79 dB share of the 82 dB budget.
Runs on	FastDSP
Hardware used	H9, H22, H23
Timing	Microseconds

Status	Designed

A6 Feed-forward ANC

Purpose (plain)	Use the outside mic to cancel rumble before it leaks into the cup.
What it does	Filter W(z) per ear. Phase 1: fixed, designed in SigmaStudio from the measured acoustic paths of our earmuff. Phase 2: adaptive (A8).
Runs on	FastDSP low-latency path
Hardware used	H2, H9
Timing	Must beat ≈ 0.3–0.6 ms; FastDSP up to 768 kHz (VERIFIED)
Status	Designed; end-to-end delay on the eval board UNVERIFIED

A7 Feedback ANC

Purpose (plain)	Catch leftover low-frequency noise at the ear.
What it does	Fixed compensator on Mic 3L / 3R, kept below the voice band for stability and so it does not attack the radio voice.
Runs on	FastDSP
Hardware used	H3, H9
Timing	Sub-millisecond loop
Status	Designed

A8 Adaptation (phase 2)

Purpose (plain)	Re-tune the anti-noise as the noise or fit changes.
What it does	FxLMS on the STM32: mic data over the DSP serial port (I²S/TDM via push-pull shifter H12b), new coefficients written over I²C every few ms. $\hat{S}(z)$ measured at start-up. FROZEN during impulses (A1/A4) and while the radio speaks (RADIO mode).
Runs on	STM32
Hardware used	H9, H11, H12b
Timing	Not in the audio path
Status	Designed; NLMS in SigmaStudio only forum-reported (UNVERIFIED), so the STM32 route is the plan

A9 Playback removal

Purpose (plain)	Stop the ANC from cancelling the hear-through we play.
What it does	Mic 3 minus (hear-through filtered by $\hat{S}(z)$). The radio voice joins after the DSP, so instead adaptation freezes in RADIO mode.

Runs on	SigmaDSP / STM32
Hardware used	H9
Timing	Per sample
Status	Designed

A10 Mixer

Purpose (plain)	Combine anti-noise and hear-through for each ear.
What it does	Mode-dependent weights (section 7).
Runs on	FastDSP
Hardware used	H9
Timing	Per sample
Status	Designed

A11 Output limiter

Purpose (plain)	Last guard inside the chip, also against bugs or howl.
What it does	Brick-wall limiter on each DAC output at the 79 dB share.
Runs on	FastDSP
Hardware used	H9
Timing	Per sample
Status	Designed

The 82 dB rule, end to end

Path	Limited where	Ceiling at the ear
ADAU output (anti-noise + hear-through)	A5 + A11	≤ 79 dB
Incoming radio voice	Jetson R5	≤ 79 dB
Sum at the speaker	—	≤ 82 dB
Amplifier maximum volume (H13)	Set once at calibration, locked by the STM32	—

Calibration (once per headset)

1. Fit the headset on the mannequin (H22); its ear mic on UMC input 4.
2. 1 kHz tone at full scale through the ADAU path; set H13 so the ear mic reads 79 dB (sound level meter H23). Lock.
3. Repeat through the radio path with R5 at full scale.
4. Measure $\hat{S}(z)$ per ear (for A6 design and A8).

5. Record passive attenuation, system off.

> **WHY NO NEURAL NETWORK IN LANE A** Apple runs ML in the ear path on its custom H2 chip (VERIFIED), so it is not physically impossible. No DSP available to us runs a neural network with a sub-millisecond path. Lane A's intelligence comes from Lane B choosing the mode.

7. Lane A modes (adaptive transparency)

> **IN PLAIN WORDS** The headset changes behaviour by itself. On patrol you hear everything a little louder than normal. When shooting starts, loud sounds are clamped harder. In a vehicle, engine rumble is cancelled and outside sound lowered. When a radio message arrives, the surroundings get quieter. With a dead battery it is still an earmuff.

Mode	When	Hear-through	ANC	Who switches
AWARE (default)	No sustained noise, no impulses	On; quiet sounds up to +12 dB	Gentle, low-frequency	Jetson classifier (B5)
COMBAT	Impulses	Harder compression; ceiling −6 dB; crack guard most sensitive	On; adaptation frozen per impulse	A4 instantly; B5 holds until ≈ 10 s quiet
VEHICLE	Loud, steady, low-frequency	−10 dB	Maximum	B5
RADIO	Incoming radio voice	Ducked −10 dB	On; adaptation frozen	Jetson (R1) → STM32
PASSIVE	No power, or ANC button	Off	Off	Physics / STM32 button

Borrowed from	Feature	Here
3M PELTOR ComTac VIII	Level-dependent hear-through, 82 dB ceiling, boom on the left	A2, A3, A5, AWARE
INVISIO X7	Passive-first protection rated with power off; body-conducted voice pickup	Section 5, PASSIVE; Mic 4 in B7
Apple AirPods Pro	Adaptive Transparency / Adaptive Audio	Automatic modes; ANC blended with hear-through
Ours	Crack guard; adaptation freeze; AI classifier steering the DSP	A4, A8, B5

8. Lane B — incoming radio voice: steps R1–R5

> **IN PLAIN WORDS** The teammate's voice arrives from the radio, is cleaned of hiss and levelled on the Jetson, and is mixed into both speakers at the amplifier, after the DSP. At the same moment Lane A switches to RADIO mode and quietens the surroundings.

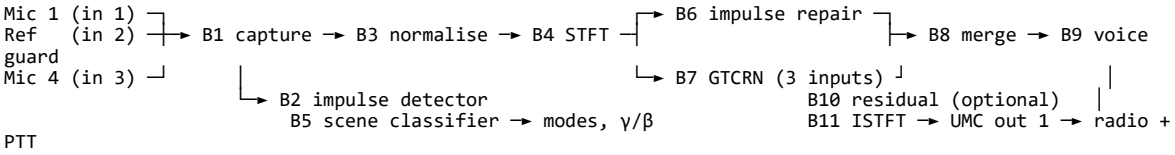
```
Radio speaker → H19 (isolated) → UMC in 4 → R1 → R2 → R3 → R4 → R5 → UMC out 2 → H13 amp (+ ADAU) → speakers
                                                  └→ voice present → UART → STM32 → RADIO mode
```

Step	What it does	Status
R1 Capture + voice detect	UMC in 4 at 48 kHz; energy + voicing detector; sends RADIO / end-of-RADIO over UART	To write
R2 Hiss / static removal	Noise gate and spectral subtraction of channel hiss and squelch tail	To write
R3 Automatic gain	Levels quiet and loud transmissions	To write
R4 Speech EQ	Emphasis at 1–4 kHz; small low-frequency lift to offset the feedback ANC	To write
R5 Limiter	Caps the radio voice at its 79 dB share	To write

Classical first. A 1-input model from the same training framework can replace R2 later if the radio channel proves noisy. If the teammate also wears ZEIT, the voice arrives already cleaned and R2–R5 mostly level and protect it.

9. Lane B — outgoing voice: steps B1–B11

> **IN PLAIN WORDS** The voice is captured by the boom mic (clear but full of gunfire) and the throat mic (muffled but almost free of gunfire); the reference mic records what the noise alone sounds like. Every 16 ms the AI compares all three and keeps only the voice. A fast detector warns when a gunshot starts so a repair step can patch it. The cleaned voice reaches the radio ≈ 70–85 ms after it was spoken.



B1 Capture and resample

Purpose (plain)	Read the three AI mics and convert to the model's rate.
What it does	UMC404HD at 48 kHz, 768-sample blocks; stateful 3:1 FIR to 16 kHz; locked channel order.
Runs on	Jetson CPU
Hardware used	H6, H14, H15
Timing	≈ 16 ms buffer + ≈ 1 ms filter
Status	Resampler MEASURED (1 kHz gain 1.000); runtime moves from 2 to 3 channels

B2 Impulse detector

Purpose (plain)	Spot a crack or blast the moment it appears.
What it does	Peak / kurtosis / high-frequency test on the reference at 48 kHz; timestamps impulses; arms B6.
Runs on	Jetson CPU
Hardware used	H14, H15
Timing	Crack known ≈ 18 ms after, ≈ 25 ms before the blast
Status	To write

B3 Robust normalisation

Purpose (plain)	Steady level without a gunshot distorting the scale.
What it does	90th-percentile or fractional-moment scaling, never RMS; identical in training and runtime.
Runs on	Jetson CPU
Hardware used	H15

Timing	Negligible
Status	MEASURED proof on 44 real events: RMS drifts 18×, FLOM 5.7×. To implement in both

B4 STFT + ERB

Purpose (plain)	Turn sound into a picture of frequencies over time.
What it does	512 FFT, 256 hop, √Hann; 65 low bins kept, 192 high bins → 64 ERB bands.
Runs on	Jetson (inside the model graph)
Hardware used	H15
Timing	32 ms window
Status	Built

B5 Scene classifier

Purpose (plain)	Decide quiet, steady noise or gunfire; set modes for both lanes.
What it does	Kurtosis + stationarity + level over ≈ 0.5 s → AWARE / COMBAT / VEHICLE; γ and β for B6; UART to STM32.
Runs on	Jetson CPU
Hardware used	H15, H11
Timing	≈ 30–60 ms, not safety-critical
Status	To write

B6 Impulse repair (BMRI)

Purpose (plain)	Patch the short moment a crack destroys.
What it does	AR interpolation on frames flagged by B2, scoped to short cracks; long blasts are conceal-and-recover, measured by IRT.
Runs on	Jetson CPU
Hardware used	H15
Timing	Within the 16 ms hop
Status	To write

B7 GTCRN, 3 inputs

Purpose (plain)	The AI: keep the voice, remove the noise.
What it does	Streaming GTCRN on boom + reference + throat; complex ratio mask on the boom mic; TensorRT FP16 then INT8.
Runs on	Jetson GPU

Hardware used	H15
Timing	Must finish in 16 ms; 2-input model 3.9 ms on a laptop CPU (MEASURED); Jetson UNVERIFIED
Status	MEASURED: 25,109 parameters; streaming and ONNX parity ≤ 6×10^{-8}. Data loader for 3 inputs to implement

B8 Merge

Purpose (plain)	Combine the AI output with repaired frames.
What it does	Crossfade to B6 inside impulse windows only.
Runs on	Jetson CPU
Hardware used	H15
Timing	Negligible
Status	To write

B9 Voice guard

Purpose (plain)	Cleaning must never remove the voice.
What it does	Restores voice bands if the output falls below what the throat envelope says was spoken.
Runs on	Jetson CPU
Hardware used	H15, H4
Timing	Negligible
Status	To write

B10 Residual LMS (optional)

Purpose (plain)	Last clean-up of slow hum.
What it does	Classical LMS on the reference; kept only if it measurably helps.
Runs on	Jetson CPU
Hardware used	H15
Timing	Negligible
Status	Optional

B11 ISTFT and radio output

Purpose (plain)	Back to sound, out to the radio.

What it does	Overlap-add at 16 kHz → 48 kHz → UMC out 1 → H19 → radio mic in. PTT is handled by the STM32, not the Jetson: the button is read on an STM32 input and the STM32 drives the H19 optocoupler, then reports PTT on/off to the Jetson over UART (the Jetson header pins supply only ≈ 2–3 mA, too little for the optocoupler — VERIFIED). PTT, not VOX (VOX would transmit gunshots).
Runs on	Jetson → H14 → H19 → H18
Hardware used	H1-G, H11, H14, H15, H18, H19
Timing	≈ 16 ms output buffer
Status	Streaming reconstruction MEASURED 4.8×10^{-7}; radio side to build

AUX-IVA REMOVED With 2–3 microphones and five or more sources the separation is underdetermined; the throat mic gives the network a noise-immune view of the voice more cheaply.

10. The neural network: 3-input GTCRN

> **IN PLAIN WORDS** GTCRN is a very small neural network — about 25,000 numbers to learn — designed to clean speech with little computing power. It sees three microphones. The throat mic is the key addition: during a gunshot the boom mic may hold almost no usable voice, but the throat mic still carries the rhythm and pitch of the words.

Property	Value
Architecture	GTCRN (Rong et al., ICASSP 2024), re-implemented, multi-channel
Inputs (locked)	0 = Mic 1 boom · 1 = reference · 2 = Mic 4 throat
Features	Magnitude, real, imaginary per input (9 channels) → ERB 129 bins → subband unfolding (27 channels)
Output	Complex ratio mask on input 0
Structure	ERB → SFE → 2 conv blocks → 3 grouped temporal conv blocks (dilation 1, 2, 5) with temporal attention → 2 dual-path grouped GRUs → mirrored decoder
Parameters	25,109 (3 inputs) · 24,389 (2) · 23,669 (1; paper 23.7 K) — MEASURED
Audio	16 kHz, 512 FFT, 256 hop — LOCKED
Streaming state	Conv cache (1,6,16,10,33); attention GRU (6,1,16); time GRU (4,33,8)
Deployment	PyTorch → streaming ONNX (parity MEASURED) → TensorRT FP16 / INT8 on the Jetson

Throat dropout

In training, input 2 is zeroed for ≈ 30 % of examples, so the same model works with or without the throat mic — no retraining if that part fails or arrives late. Throat-channel training data: real pairs from TAPS and Vibravox (VERIFIED to exist), and a simulated throat channel for other speech, checked against held-out real pairs.

What must stay fixed between training and hardware

Link	Hardware	Training	Status
Inputs and order	UMC in 1 / 2 / 3	Inputs 0 / 1 / 2	Locked — label cables
Sample rate	48 kHz capture	16 kHz model	Resampler MEASURED
Frame / hop	768 @ 48 kHz	256 @ 16 kHz	Parity MEASURED
Reference clipping	Overload at 135 dB	Clipping augmentation	Built
Per-mic gain	H6 per channel	Random per-channel gain	Input 1 built; input 2 to add
Voice in reference	Head between boom and Mic 2R	Leak −45 … −25 dB	Built; voice-leak test
Throat channel	H4	Real + simulated + dropout	To implement

Link	Hardware	Training	Status
Normalisation	B3 runtime	Same B3 in T2	To implement
Precision	TensorRT FP16 / INT8	Compared with FP32	Tools built

11. Training and deployment: steps T1–T8

ID	Step	What happens	Code
T1	Index	Scan the data drive once: speech, noise, room responses, splits	build_index.py — built
T2	Mix	Speech + noise at SNR −15 … +20 dB; simulated reference; throat channel; dropout; B3	Built for 3 inputs (throat simulated, 30 % dropout); real throat pairs + B3 to add
T3	Augment	α-stable bursts incl. α < 0.5, clipping, room responses, 20 % Lombard, per-channel gain	Built
T4	Train	30·complex + 70·magnitude (compressed) + SI-SNR + multi-resolution STFT; AdamW; cosine; early stop	train.py — built
T5	Evaluate	SNR out, ΔSNR, SI-SNR, PESQ, STOI per input-SNR bucket; IRT, WLPS; after a vocoder too	evaluate.py — built; vocoder to add
T6	Prune (optional)	Magnitude pruning, fine-tune, re-evaluate	prune.py — built
T7	Quantise	INT8 QDQ calibrated on real noisy recordings	quantize_onnx.py — built
T8	Export + deploy	Streaming ONNX with parity checks → TensorRT → live runner	export_onnx.py, jetson/ — built; 3-input export MEASURED

ORDER A single-channel sanity run on VoiceBank-DEMAND must reach PESQ near the paper's 2.87 before the full 3-input run.

12. How the lanes connect

Link	From	To	Carries
Shared mic	Mic 2R via H6 (two buffered copies)	ADAU ADC 2 and UMC in 2	The noise reference, neither board loading the other
Modes down	Jetson B5 / R1 (UART)	STM32 → ADAU (I²C)	AWARE / COMBAT / VEHICLE / RADIO
Status up	STM32 → UART	Jetson	PTT on/off, crack events, mode acknowledgement
Incoming voice	Jetson R5 → UMC out 2	H13 amp, summed with ADAU	Cleaned teammate voice to both ears
Outgoing voice	Jetson B11 → UMC out 1	H19 → radio mic in	Cleaned wearer voice
PTT	Button → STM32 input; STM32 → H19 optocoupler	Radio PTT	Transmit on / off

UART protocol (Jetson ↔ STM32)

Direction	Message	Meaning
Jetson → STM32	MODE AWARE / COMBAT / VEHICLE	Scene mode from B5
Jetson → STM32	RADIO 1 / RADIO 0	Incoming radio voice present / ended (R1)
STM32 → Jetson	PTT 1 / PTT 0	Push-to-talk pressed / released
STM32 → Jetson	CRACK	Crack guard fired
STM32 → Jetson	ACK <mode>	Mode applied to the DSP
Both	PING / PONG	Link check every second; if the Jetson is silent the STM32 keeps the last mode

LINK 115200 baud, 8N1, 3.3 V, one ASCII line per message.

DESIGN RULE Nothing that protects the ear depends on the Jetson. If it crashes, Lane A keeps its last mode, the crack guard and limiters still work, and the earmuff still protects. At power-on the STM32 loads the DSP program into the ADAU (unless the eval board self-boots — check UG-1532).

13. One gunshot, millisecond by millisecond

Time	Where	What happens (rifle 30 m, supersonic, AWARE, soldier speaking — ESTIMATE)
0 ms	Rifle	Trigger; blast travels at 343 m/s; the bullet travels faster
36.5 ms	Bullet, ~3 m	Closest approach; the crack forms near the soldier
45.2 ms	Mic 2L, 2R	Crack arrives; Mic 2R likely clips
45.2 ms + µs	Lane A	A1 + A4: hear-through −20…30 dB both ears; A8 frozen; COMBAT
≈ 61–63 ms	Lane B	Crack's block reaches the Jetson; B2 flags; B6 armed; B5 confirms COMBAT
87.5 ms	Head	Blast. Earmuff (+ plug) takes the peak; hear-through already ducked and capped
87.5–300 ms	Lane B	GTCRN masks the boom mic guided by the throat mic; B6 patches short gaps
≈ 350–450 ms	Lane A	Crack guard releases; adaptation resumes
+10 s quiet	Both	Back to AWARE

14. Latency

Lane A

Path	Limit	Design
ANC (A6)	≈ 0.3–0.6 ms acoustic path	FastDSP up to 768 kHz (path VERIFIED; end-to-end UNVERIFIED)
Hear-through	≤ 5–6 ms (hearing-aid studies, VERIFIED)	Well under 1 ms (ESTIMATE)
Crack guard	Before the blast, ≈ 42 ms after the crack	A few samples (ESTIMATE)
Mode change	Hundreds of ms acceptable	≈ 30–60 ms (ESTIMATE)

Lane B — mouth to radio

Step	Delay	Status
Interface input buffer	≈ 16 ms	ESTIMATE
Resampling down + up	≈ 2 ms	ESTIMATE
STFT window	32 ms	By design
Compute per frame (≤ 16 ms)	≈ 2–6 ms	3.9 ms laptop CPU MEASURED; Jetson UNVERIFIED
Interface output buffer	≈ 16 ms	ESTIMATE
Total	≈ 70–85 ms	ESTIMATE
ITU-T G.114 one-way limit	≤ 150 ms	VERIFIED

The incoming radio voice (R1–R5) adds ≈ 20–40 ms (ESTIMATE), unnoticeable when listening to a radio.

15. Failure behaviour

Failure	Wearer gets	Listener gets
Power lost	Passive earmuff protection	Nothing (radio off)
Jetson crash	Lane A in its last mode; crack guard and limiters work	Nothing until restart
ADAU fault	Passive protection; radio voice still reaches the ears via H13	Unaffected
Mic 2 clipped	Crack guard + frozen adaptation	Model trained on clipped references
Throat mic missing	Unaffected	Model continues with the throat input at zero (trained with 30 % dropout)
Mic 3 failed	ANC in that ear stops; hear-through works	Unaffected

16. Prototype versus fielded unit

Bench prototype	Fielded unit
ADAU1787 eval board + STM32 Nucleo	ADAU1787 chip + small STM32 on one PCB inside the right earcup
UMC404HD USB interface	Audio codec on the Jetson carrier over I^2S
Jetson developer kit	Jetson Orin Nano module on a small carrier in a chest-mounted unit (same form factor as tactical headset control units)
Mains / power bank	Radio or vest battery; Jetson in its 7 W mode

17. Proven versus designed

Proven

Claim	Evidence
RMS normalisation fails on gunshots	MEASURED on 44 real events: RMS 18×, FLOM 5.7× within an event
IRT exposes damage PESQ / STOI hide	MEASURED: the 1–2 kHz band takes 464 ms to recover
Our own impulsive recordings	Range trip 7 Sep 2026: 15 takes, 213 events, 44 impulses (air gun, uncalibrated)
Model and deployment path work	MEASURED: 25,109 parameters; parity ≤ 6×10^{-8}; 3.9 ms/frame (laptop CPU, 2-input); 12/12 self-test
Consumer ANC excludes gunfire	VERIFIED: Apple documentation

Designed

Claim	Verified so far
Two lanes on one headset	ADAU1787 ANC-oriented low-latency path; Jetson TensorRT platform
Both-ear hybrid ANC on one DSP	4 ADC / 2 DAC matches a stereo hybrid-ANC headset
Crack guard	Physics; DSP detector capability
3-input AI with a throat mic	Body-conducted voice rejects airborne noise; paired datasets exist
Peltor-style hear-through, automatic modes	82 dB practice; limiter blocks native to the DSP

18. Gates and open checks

Before ordering

#	Check	Parts
1	UG-1532: built-in USB? I/O voltage / level shifting? self-boot? serial-port pins? analog input connectors?	H9–H12
2	ADAU1787 FastDSP capacity for two ANC paths	H9
3	IM73A135 as a ready breakout in India	H2
4	Earmuff price; cup depth fits a 40 mm speaker + inside mic	H1
5	PMR446: WPC ETA, K-type jack, pinout	H18, H19
6	Written Jetson quotation	H15

After arrival

1. UMC404HD records 4 channels on the Jetson.
2. 3-input model builds as a TensorRT engine; GRUs convert; frame time < 16 ms.
3. Voice-leak test: reference ≥ 25 dB below Mic 1.
4. Hum check: grounding first, isolation only if needed.
5. SigmaStudio: hear-through + limiters, then fixed ANC, then adaptation.
6. Throat-mic spectra: simulated vs real on held-out pairs.

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ZEIT System Architecture · write changes here · reference step / item IDs (e.g. A4, B7, H9) so both documents stay aligned

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